

## Question Bank — Week 2

Complexity and the Array · control flow, functions, errors (Lecture 02, Lab 02) —  
Answers

DSA27 · Data Structures and Algorithms

Fall 2026

**Questions:** *week02-questions.md*. Commit to your own answer before reading one here.

### Part A — Multiple choice

| Q   | Ans | Q   | Ans | Q   | Ans | Q   | Ans |
|-----|-----|-----|-----|-----|-----|-----|-----|
| M01 | b   | M08 | a   | M15 | d   | M22 | c   |
| M02 | c   | M09 | b   | M16 | b   | M23 | d   |
| M03 | a   | M10 | c   | M17 | a   | M24 | a   |
| M04 | d   | M11 | d   | M18 | c   | M25 | b   |
| M05 | a   | M12 | a   | M19 | d   | M26 | c   |
| M06 | c   | M13 | b   | M20 | a   |     |     |
| M07 | d   | M14 | c   | M21 | b   |     |     |

**W2-M01 — b.** Keep the fastest-growing term, drop its constant:  $3n^2 \rightarrow n^2$ . It is also  $O(n^3)$ , but not  $\Theta(n^3)$  (c) —  $\Theta$  is a tight bound.

**W2-M02 — c.**  $n \leq 1 \cdot n^2$  for all  $n \geq 1$ .  $O$  is an *upper* bound, so a slower-growing function is  $O$  of a faster one. (a) and (d) put a faster function under a slower one; (b) claims  $n$  grows at least as fast as  $n^2$ .

**W2-M03 — a.** Constants **you choose**, and only from some  $n_0$  on. (b) has no constant and no  $n_0$ ; (c) is equality, not a bound; (d) is the definition of  $\Omega$ .

**W2-M04 — d.** Worst case: the target is last or absent, so every element is compared once.

**W2-M05 — a.**  $(1 + 2 + \dots + n)/n = (n + 1)/2$ . Still  $O(n)$  — half is a constant.

**W2-M06 — c.** Halving until 1 takes about  $\log_2 n$  steps.  $\Theta(n/2)$  (d) would be subtracting, not dividing — and it is just  $\Theta(n)$  anyway.

**W2-M07 — d.**  $0 + 1 + \dots + (n - 1) = n(n - 1)/2$ . The answer must be a function of  $n$  only, so (c) is not an answer.

**W2-M08 — a.** The inner loop is a constant 5:  $5n = \Theta(n)$ . Nested loops multiply their *iteration counts*, not their nesting depth.

**W2-M09 — b.** A list is an array; `in` checks items one by one. Sets and dicts are the hash tables (a).

**W2-M10 — c.** `append` writes at the end — nothing moves — and the occasional resize averages out (Week 4). The other three shift or search:  $O(n)$ .

**W2-M11 — d.** Address arithmetic. It has nothing to do with order (a) or caching (b).

**W2-M12 — a.** Slots 2, 3, 4, 5 move right:  $\text{size} - \text{index} = 6 - 2 = 4$ .

**W2-M13 — b.** Moving right, start at the right, or you overwrite values before you move them (see W2-S2).

**W2-M14 — c.** The course Array has no negative indices, like C.

**W2-M15 — d.** `len` is the capacity, fixed at creation. Tracking how many slots are in use is your job.

**W2-M16 — b.**  $r \times \text{cols} + c = 2 \times 4 + 1 = 9$ .

**W2-M17 — a.** Two indices and one temporary for the swap, whatever `n` is.

**W2-M18 — c.** A structure may not store its data in a Python `list`. (a) and (b) build on the student's own structures — allowed; (d) is a list used as a return value — the interface, allowed.

**W2-M19 — d.** A block of references plus a size and a capacity; it grows by copying into a bigger block.

**W2-M20 — a.** Sequence adds:  $O(n + n^2) = O(n^2)$ .

**W2-M21 — b.** Each iteration hides **two**  $O(n)$  operations: the slice copies up to `n` items and `in` searches them. `n` iterations of  $O(n)$  work.

**W2-M22 — c.** Think of it as “no break”.

**W2-M23 — d.** The default list is created **once**, when `def` runs, and shared by every call that uses it.

**W2-M24 — a.** `and` binds tighter: `0` or `('x' and '')`. `'x'` and `''` gives `''` (the last value looked at); `0` or `''` gives `''`.

**W2-M25 — b.** Start 2, step 3, stop **before** 11: 2, 5, 8.

**W2-M26 — c.** `return 1` is decided, then `finally` runs (printing `bye`) before the function actually returns; then `print` shows 1.

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## Part B — Short answer and essay

### W2-E1 (4)

- **Lower-order terms** stop mattering as `n` grows: in  $\frac{1}{2}n^2 - \frac{1}{2}n$ , the  $n^2$  term is 99.9% of the total by `n` = 1,000.

- **Constant factors** are not properties of the algorithm: they change with the machine, the language and how you count steps. What stays the same everywhere is the **shape** of growth.
- **Misleading when n is small**, or constants are huge: a  $1000n$  algorithm is worse than an  $n^2$  one for every  $n < 1,000$ ; linear search beats binary search on tiny arrays.
- **Measuring** (viz. `complexity.measure`) shows the constants and the crossover point that Big-O hides; counting shows the shape that one machine's timings cannot prove. You need both.

#### W2-E2 (4)

- $f = O(g)$ : there exist  $c > 0, n_0$  with  $f(n) \leq c g(n)$  for all  $n \geq n_0$  — an upper bound.
- $f = \Omega(g)$ : there exist  $c > 0, n_0$  with  $f(n) \geq c g(n)$  for all  $n \geq n_0$  — a lower bound.
- $f = \Theta(g)$ : both —  $c_1 g(n) \leq f(n) \leq c_2 g(n)$  eventually — a tight bound.
- For  $3n^2 + 2n$ : it is  $O(n^2)$ ,  $\Omega(n^2)$ ,  $\Theta(n^2)$ ; it is also  $\Omega(n)$ . **True but uninformative:**  $3n^2 + 2n = O(n^3)$  (or  $O(2^n)$ ).

#### W2-E3 (3)

- **Best:** the target is the first element — 1 comparison,  $O(1)$ .
- **Worst:** last or absent —  $n$  comparisons,  $O(n)$ .
- **Average** (present, uniformly placed):  $(n + 1)/2$ , still  $O(n)$  — and the average needs a stated assumption about the inputs.
- "Linear search is  $O(n)$ " means the **worst case**: it is a guarantee that holds for every input of size  $n$ , needs no assumption, and is what you plan capacity around.

#### W2-E4 (4)

- An array is contiguous, equal-sized slots, so the address of `a[i]` is `base + i × size`: one multiplication and one addition, independent of  $n \rightarrow O(1)$ .
- Inserting at index  $i$  in an array of size  $n$  needs every element from  $i$  to the end moved one place right:  $n - i$  moves,  $n$  in the worst case  $\rightarrow O(n)$ .
- **Fixed size:** the memory after the block may belong to something else, so the array cannot grow in place. When it is full, you must allocate a **new, larger** array and **copy** every element across:  $O(n)$  for that operation.
- (Doubling the capacity each time makes appends  $O(1)$  amortised — Week 4.)

#### W2-E5 (4)

- **Rule:** a data structure in `dsa/` stores its data only in the course `Array`, in node objects, or in another structure the student has built — never in a Python `list`, `dict`, `set` or `deque`.
- **Reason:** those built-ins hide exactly the costs the course teaches — one short call like `insert(0, x)`, `pop(0)` or `x in values` is an  $O(n)$  algorithm someone else wrote. Building on a bare array makes every cost visible and yours.
- **Still allowed:** lists as function inputs and return values; lists in tests; plain local variables and tuples.
- **Examples**, any two: `Stack` on the student's `DynamicArray`; `CircularQueue` on an `Array`; `ChainingHashMap` as an `Array` of `Entry` chains; `Graph` on the student's hash map and `DynamicArrays`; `MinHeap` on a `DynamicArray`.

## W2-E6 (3)

|                            | Grows?              | Holds                      | Hides the cost?                       |
|----------------------------|---------------------|----------------------------|---------------------------------------|
| <code>list</code>          | yes (dynamic array) | references to any objects  | yes                                   |
| <code>array.array</code>   | yes                 | packed numbers of one type | yes                                   |
| <code>numpy.ndarray</code> | no                  | packed numbers of one type | yes — whole-array operations run in C |
| <code>ctypes array</code>  | no                  | C values or references     | no — index and assign only            |

A `ctypes` array of object references is a genuine fixed-size C array that can hold any Python object. The course wraps it, adds bounds checking, and offers nothing else — so every other operation must be written, and priced, by the student.

## Part C — Complexity analysis

**W2-K1** —  $\Theta(n^2)$ . The nested loops do  $n \times n$  iterations of  $O(1)$  work; the following loop adds  $n$ . Sequence adds:  $n^2 + n = \Theta(n^2)$ .

**W2-K2** —  $\Theta(\log n)$ .  $i$  takes the values 1, 2, 4, 8, ...; it reaches  $n$  after about  $\log_2 n$  doublings. Doubling up to  $n$  is the mirror of halving down from  $n$ .

**W2-K3** —  $\Theta(n \log n)$ . The outer loop runs  $n$  times; each time the inner loop halves  $j$  from  $n$  down to 1, about  $\log_2 n$  iterations. Loops multiply. (Counting: for  $n = 1,024$  the inner body runs exactly 10,240 times.)

**W2-K4** — **best**  $\Theta(1)$ , **worst**  $\Theta(n^2)$ . Best: `arr[0] == arr[1]`, found on the first comparison. Worst: no duplicates, so every pair is compared:  $n(n-1)/2$  comparisons.

**W2-K5** —  $\Theta(n^2)$ . The loop body looks like one step, but `values[i:]` **copies**  $n-i$  items. Total copying:  $n + (n-1) + \dots + 1 = n(n+1)/2$ .

**W2-K6** —  $\Theta(n^2)$ . The middle loop is a constant 100. Total  $100 \times (0 + 1 + \dots + (n-1)) = 50n(n-1)$ . A large constant is still a constant. (For  $n = 10$ , count is 4,500.)

## Part D — Proofs

**W2-P1.** Choose  $c = 3$ ,  $n_0 = 10$ . For every  $n \geq 10$ ,  $10 \leq n$ , so  $2n + 10 \leq 2n + n = 3n$ . Hence  $2n + 10 \leq 3 \cdot n$  for all  $n \geq 10$ , and  $2n + 10 = O(n)$ . ■ (Other valid choices exist, e.g.  $c = 12$ ,  $n_0 = 1$ .)

**W2-P2.** Suppose, for contradiction, that  $n^2 = O(n)$ : there are  $c > 0$  and  $n_0$  with  $n^2 \leq c n$  for all  $n \geq n_0$ . Dividing by  $n > 0$  gives  $n \leq c$  for all  $n \geq n_0$ . But take  $n = \max(n_0, \lceil c \rceil + 1)$ : then  $n \geq n_0$  and  $n > c$  — a contradiction. So no such  $c$  exists, and  $n^2$  is not  $O(n)$ . ■

**W2-P3.** Need  $c_1, c_2 > 0$  and  $n_0$  with  $c_1 n^2 \leq 4n^2 + n \leq c_2 n^2$  for all  $n \geq n_0$ .

- **Lower bound:**  $4n^2 + n \geq 4n^2$  for all  $n \geq 0$ , so  $c_1 = 4$  works:  $4n^2 + n = \Omega(n^2)$ .
- **Upper bound:** for  $n \geq 1$ ,  $n \leq n^2$ , so  $4n^2 + n \leq 5n^2$ ;  $c_2 = 5$ :  $4n^2 + n = O(n^2)$ .
- With  $n_0 = 1$  both hold, so  $4n^2 + n = \Theta(n^2)$ . ■

## Part E — Array state

### W2-S1.

(a) Shift 50, 40, 30, 20 one place right (4 moves, starting from 50), then write 15 at index 1:

```
[10, 15, 20, 30, 40, 50, _]      size 6
```

(b) Remove index 3 (the value 30): shift 40 and 50 one place left (2 moves), then clear the slot that is no longer used:

```
[10, 15, 20, 40, 50, _, _]      size 5, returned 30
```

### W2-S2.

```
start          [a, b, c, d, _]
i = 1: arr[2] = arr[1]  -> [a, b, b, d, _]    c is overwritten before it moved
i = 2: arr[3] = arr[2]  -> [a, b, b, b, _]
i = 3: arr[4] = arr[3]  -> [a, b, b, b, b]
arr[1] = "X"          -> [a, X, b, b, b]
```

Copying forwards overwrites each value before it has been moved, so one value (b) is smeared across the rest and c and d are lost. **Fix:** move from the end backwards:

```
for i in range(size, index, -1):
    arr[i] = arr[i - 1]
arr[index] = value
```

### W2-S3.

```
start          [1, 2, 3, 4, 5, 6]
reverse the first 2 (0..1)  [2, 1, 3, 4, 5, 6]
reverse the rest (2..5)    [2, 1, 6, 5, 4, 3]
reverse the whole (0..5)   [3, 4, 5, 6, 1, 2]
```

Each reversal is a two-pointer swap:  $O(n)$  time in total,  $O(1)$  extra space.

### W2-S4.

(a) Value **7**, at flat index  $1 \times 4 + 3 = 7$  (the values are the indices, because they were stored row by row).

(b) Row  $10 // 4 = 2$ , column  $10 \bmod 4 = 2$ : **(2, 2)**.

(c) Transposing moves (1, 3) to (3, 1) in a  $4 \times 3$  matrix: flat index  $3 \times 3 + 1 = \$ 10$ .

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## Part F — Trace the code

### W2-T1

```
11 13 done
```

The `else` belongs to the inner `for`: it prints `n` only when no divisor was found, i.e. `n` is prime. 10, 12, 14 and 15 hit `break`.

### W2-T2

```
5 -4 0 3 ?
```

"sum" matches `["sum", *rest]` with `rest = []`, so it returns 0. "add 1" has only two words, so it fails the `add` pattern (three words) and falls to `_`.

### W2-T3

```
A B D E 5
A C E 0
```

`finally` always runs, **before** the returned value reaches `print`. With 0, the division raises, so `B` is skipped and `except` prints `C`; `else` runs only when the `try` did not raise.

### W2-T4

```
[1, 2] 2 20
```

`values.append(n)` mutates the caller's list. `n = n * 10` rebinds a local name — `k` is untouched. `values = [0]` rebinds the local name too — `data` keeps `[1, 2]`.

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## Part G — Find and fix the bug

**W2-B1.** `range(2, int(n ** 0.5))` stops **before** the square root, so a perfect square of a prime is never tested against its root: `is_prime(9)`, `is_prime(25)` and `is_prime(49)` return `True`. **Fix:** `range(2, int(n ** 0.5) + 1)` — or, avoiding floats, `d = 2; while d * d <= n:`.

**W2-B2.** The `else: return -1` is inside the loop, so the function gives up after checking **only** `arr[0]`. `find([4, 7], 7)` returns `-1`. **Fix:** move `return -1` after the loop, and delete the `else`.

**W2-B3.** The loop runs one step too far: for `i = size - 1` it reads `arr[size]`, one past the used part — an `IndexError` when the array is full (`size == len(arr)`), and a silent read of a free slot otherwise. The slot that is no longer used is also never cleared. **Fix:**

```
for i in range(index, size - 1):
    arr[i] = arr[i + 1]
arr[size - 1] = None
```

## Part H — Write the code

### W2-C1

```
def count_occurrences(arr, target, size=None):
    n = len(arr) if size is None else size
    count = 0
    for i in range(n):
        if arr[i] == target:
            count += 1
    return count
```

**W2-C2** — a *read* index and a *write* index.

```
def remove_all(arr, size, target):
    write = 0                                # next slot to keep a value in
    for read in range(size):
        if arr[read] != target:
            arr[write] = arr[read]
            write += 1
    for i in range(write, size):              # clear the freed tail
        arr[i] = None
    return write
```

Each element is read once and written at most once:  $O(n)$ ,  $O(1)$  extra. Calling `remove_at` for each occurrence shifts up to  $n$  elements **per removal**; with  $k$  occurrences that is  $O(kn)$ , which is  $O(n^2)$  when  $k$  is proportional to  $n$ .

**W2-C3** — two indices moving towards each other.

```
def two_sum_sorted(arr, target):
    lo, hi = 0, len(arr) - 1
    while lo < hi:
        total = arr[lo] + arr[hi]
        if total == target:
            return (lo, hi)
        if total < target:
            lo += 1                                # need a bigger sum
        else:
            hi -= 1                                # need a smaller sum
    return None
```

**Why it is correct:** if  $\text{arr}[\text{lo}] + \text{arr}[\text{hi}]$  is too small, then  $\text{arr}[\text{lo}]$  plus *any* element at or before  $\text{hi}$  is also too small (the array is sorted), so  $\text{arr}[\text{lo}]$  can be in no solution — discard it. Symmetrically for too big. Each step discards one element, so at most  $n - 1$  steps:  $O(n)$ .

#### W2-C4

```
def prefix_sums(arr):  
    result = Array(len(arr))  
    running = 0  
    for i in range(len(arr)):  
        running += arr[i]  
        result[i] = running  
    return result
```

Then  $\text{sum}(\text{arr}[i..j]) = p[j] - p[i - 1]$  (or  $p[j]$  when  $i = 0$ ): one subtraction,  $O(1)$ , after one  $O(n)$  pass. Paying once up front to answer many questions cheaply is a trade you will see again and again.